SGIC WHITEPAPER SERIES

Visceral Adiposity, Wheat Amylase-Trypsin Inhibitors and Dietary Neu5Gc

A two-signal model of mesenteric inflammatory tone, stated by evidence tier

Martin Hamer, RN (Ret.)

Hamer Glycan Systems — hamerglycans.com

August 2026

Summary

This paper sets out what the Sialic-Glycan Inflammatory Cascade (SGIC) framework can and cannot claim about visceral adipose tissue. It makes two claims and keeps them apart deliberately.

The first claim, made at full strength, is that **removing wheat amylase-trypsin inhibitors (ATIs) and dietary Neu5Gc reduces the inflammatory tone arising from mesenteric adipose tissue**, with consequences for hepatic steatosis, insulin sensitivity and endothelial function. This claim does not depend on any change in fat mass.

The second claim, made explicitly as a hypothesis, is that **the same removals reduce visceral fat mass itself**. This paper argues that the second claim is currently unsupported, that the available human trial cannot distinguish it from simple energy restriction, and that pharmacological IL-1β blockade provides direct evidence against it. It is retained here as a falsifiable proposition with a study design attached, not as a conclusion.

Conflating the two has been the principal weakness of dietary inflammation literature. The framework is stronger, not weaker, for separating them.

1. Evidence tiers used in this paper

Every substantive claim below is labelled A, B or C. The tiers are not rhetorical hedging; they determine what a reader is entitled to conclude.

Tier	Standard of evidence	Claims carried at this tier
A	Established. Mechanism	Wheat ATIs activate TLR4 on intestinal

Tier	Standard of evidence	Claims carried at this tier
	demonstrated in animals and endpoint moved in at least one human randomised trial.	myeloid cells; ATI withdrawal improves hepatic steatosis and insulin resistance over six weeks.
B	Animal proof-of-concept. Causal demonstration in a humanised model; no human interventional data.	Dietary Neu5Gc, in the presence of anti-Neu5Gc antibodies, accelerates atherosclerosis independently of lipid levels; excess dietary Neu5Ac partially reverses it.
C	Hypothesis. Mechanistically coherent, not yet tested; stated here so it can be falsified.	Anti-Neu5Gc IgG engages FcγR and complement on ATI-primed myeloid cells within mesenteric adipose tissue, and the two signals converge there.

A reader should treat Tier A as actionable, Tier B as a reason to run the human study, and Tier C as a statement of what the author believes but has not shown.

2. The ATI arm (Tier A)

Wheat amylase-trypsin inhibitors are non-gluten protein components of wheat that survive digestion and act as ligands for Toll-like receptor 4 on intestinal myeloid cells. This is not inferred from dietary epidemiology; it was established in defined experimental systems and reproduced across laboratories.

In murine work, ATI feeding promoted the features of metabolic dysfunction-associated steatotic liver disease, insulin resistance and hepatic fibrosis, with the effect traced to TLR4 activation on myeloid cells in the gut rather than to any direct hepatic action.

2.1 The human trial, and precisely what it showed

A six-week, 1:1 randomised, controlled proof-of-concept study in patients with biopsy-proven MASLD compared an ATI-reduced (gluten-free) diet against a balanced diet following German Nutrition Society guidance. Forty-five patients were enrolled, twenty-one to intervention and twenty-four to control, with three discontinuations in each arm.

The intervention arm showed significant reductions in controlled attenuation parameter, a transient-elastography measure of hepatic steatosis; in HOMA-IR; and in body mass index.

Three points of precision matter more than the result itself. First, the steatosis endpoint was hepatic, measured by CAP, not visceral fat mass measured by imaging. Hepatic fat is diet-composition-sensitive and moves within weeks; visceral fat mass is slow and largely energy-driven. These are different tissues on different timescales and the trial speaks only to the former. Second, energy intake was not controlled, and the

trial's own authors note that the observed BMI reduction may be partly responsible for the improvement in insulin sensitivity. Third, six weeks is a proof-of-concept duration in forty-five patients; it establishes a signal worth pursuing, not an effect size worth quoting.

Cited honestly, this trial is a genuine asset. Cited as evidence that wheat withdrawal reverses visceral fat, it invites immediate and fatal correction.

3. The Neu5Gc vascular arm (Tier B)

Humans cannot synthesise N-glycolylneuraminic acid, having lost a functional CMAH gene some two to three million years ago. Neu5Gc is nonetheless taken up from mammalian dietary sources through the sialic acid salvage pathway and metabolically incorporated into the consumer's own surface glycoconjugates by the consumer's own sialyltransferases. It is not deposited; it is installed. Because essentially all humans carry circulating anti-Neu5Gc antibodies, the result is a xenoautoantigen: a foreign epitope presented on self scaffolds.

3.1 The causal demonstration

Kawanishi and colleagues addressed this directly in a doubly deficient mouse (Cmah-null crossed onto an Ldlr-null background) engineered to reproduce the human immunological situation. Animals were immunised to raise human-like anti-Neu5Gc antibodies and then fed a Neu5Gc-rich diet. Atherosclerosis was substantially increased, and the authors distinguished two contributing mechanisms: an intrinsic effect of CMAH loss itself, and an extrinsic effect requiring dietary Neu5Gc together with the antibody response.

The intrinsic–extrinsic distinction deserves emphasis, because it constrains the framework. Part of the observed vascular phenotype arose from the genetic background irrespective of diet. Only the extrinsic component is dietary, and therefore only the extrinsic component is modifiable by the intervention this framework proposes. Overstating the dietary share of that effect is an error a reviewer will find immediately.

3.2 The competition experiment

A subsequent study from the same group is arguably more useful to a dietary framework than the original. Rather than eliminating Neu5Gc, the investigators fed a five-fold excess of Neu5Ac derived from edible bird's nest alongside a Neu5Gc-rich high-fat diet. Plaque size fell significantly relative to the Neu5Gc-fed comparison, together with a reduction in necrotic core formation, and the effect held despite elevated triglyceride-rich lipoproteins.

This establishes competitive displacement as a second lever alongside avoidance. Where complete elimination of mammalian Neu5Gc is impractical — which is most real-world

circumstances — saturating the salvage pathway with Neu5Ac is a mechanistically distinct and independently testable approach.

3.3 The standing critique

The Cmah-null model has been challenged in print. Soulillou and colleagues argued in PNAS that a model combining CMAH loss with LDL-receptor loss and adjuvant-driven immunisation amplifies the xenosialitis effect well beyond anything human physiology would produce, and that extrapolation to human atherosclerosis is therefore unsafe. The original authors replied acknowledging the general limits of animal extrapolation while defending the model′s relevance to the human propensity for atherosclerotic disease.

This exchange belongs in the framework′s own literature rather than being left for a reviewer to introduce. Citing a critique of one′s central supporting study is a credibility marker, not a concession.

4. The mesenteric adipose hypothesis (Tier C)

The proposition specific to this framework is that mesenteric adipose tissue is where the two signals meet.

The model has two arms. ATIs prime TLR4 on myeloid cells within the mesenteric compartment, raising the activation threshold response of those cells. Anti-Neu5Gc IgG, encountering Neu5Gc-bearing epitopes installed on local endothelium and stromal cells, then engages Fcγ receptors and fixes complement on the same primed population. Sublytic membrane attack complex formation activates NLRP3 and drives IL-1β release. Because mesenteric lymph drains this compartment, the products of that convergence reach the liver by the portal route and the systemic circulation by the thoracic duct.

The ATI arm of this model is Tier A. The Neu5Gc arm is Tier C. The convergence itself has never been demonstrated in any tissue and is offered here as a hypothesis.

4.1 The detection problem

The most direct challenge is the reported absence of detectable Neu5Gc in human adipose tissue. That finding is real and must be addressed rather than ignored.

Guerrero-Flores and colleagues (2023) measured Neu5Ac, Neu5Gc and KDN by LC–MS/MS in adipose tissue from 156 participants of the Adventist Health Study-2: 52 vegans, 56 lacto-ovo-vegetarians and 48 non-vegetarians. Neu5Gc was not detected in any human sample from any dietary group. The same method detected Neu5Gc in adipose tissue from cow, lamb and pig and correctly failed to detect it in chicken, so the assay is competent on this matrix.

Three caveats are legitimate and are stated here rather than relied upon. First, the samples were subcutaneous aspirates from the buttock; the visceral depot with which

this paper is concerned has not been assayed. Second, the lower limit of detection was at or above 2.0 ng/mL, and the authors themselves note that Neu5Gc may be present below that threshold. Third, non-vegetarians in this cohort eat less red meat than the general population.

The framework's response is arithmetic rather than rhetorical. Neu5Gc is not stored in tissue in the manner of a lipophilic compound; it is installed into cell membranes, and adipose tissue is upwards of 85 percent triglyceride by mass, so any membrane glycan is heavily diluted when expressed per milligram of tissue. The same study reported adipose Neu5Ac in non-vegetarians at about 9.8 ng per mg. If Neu5Gc makes up between a small fraction of one percent and about one percent of total sialic acid, as in other non-malignant human tissues, expected adipose Neu5Gc is roughly 0.001 to 0.1 ng per mg. With samples of 5 to 30 mg reconstituted in 100 microlitres, that gives an extract concentration of about 0.05 to 30 ng/mL, most of which lies below the 2.0 ng/mL floor. The non-detection is therefore largely predicted by the sensitivity of the assay. It is not evidence that Neu5Gc is absent from adipose tissue; it is an unresolved measurement problem.

This is a legitimate response, but it is a response about sensitivity, and it obliges the framework to state what would count against it. The Neu5Gc arm of this model would be falsified for the visceral depot if a validated assay with a lower limit of detection at or below 0.01 percent of total sialic acid (approximately 0.001 ng per mg of tissue, the lowest concentration the model predicts), confirmed by spike recovery in the same matrix, failed to detect Neu5Gc in mesenteric or omental adipose tissue from non-vegetarian subjects with documented dietary exposure. Below that level the antigen would be too sparse to sustain antibody-mediated inflammation, and its absence would refute this arm of the hypothesis rather than merely fail to support it. Until such a measurement exists, the Neu5Gc arm remains Tier C. That classification rests on the absence of a sufficiently sensitive assay, not on a demonstrated absence of the antigen.

5. Gliadins: barrier permeability only

Gliadins are included in this framework at one point and one point only, as contributors to Gate Two of the three-gate susceptibility model — that is, as agents that increase intestinal permeability and thereby govern whether antigen and adjuvant reach the submucosal compartment together.

Gliadin-driven zonulin release and the consequent loosening of tight junctions provide a plausible mechanism for that role. What gliadins are not offered as, anywhere in this paper, is a driver of adipose tissue biology. There is no adipose-specific gliadin mechanism, and importing coeliac literature into a metabolic argument would introduce a body of evidence that points, if anything, the other way.

Stated as a permeability modifier, the gliadin claim is modest and defensible. Stated as an adiposity mechanism, it would be the weakest element in the paper and would attract disproportionate criticism to the whole.

6. Direction of causation

A tempting syllogism runs through much of the popular literature on this subject: visceral fat produces endothelial inflammation; Neu5Gc produces endothelial inflammation; therefore removing Neu5Gc reduces visceral fat. The conclusion does not follow. Two independent inputs to a common pathway do not make either one a lever on the other.

The better-supported direction is the reverse. Inflamed visceral and mesenteric adipose delivers free fatty acids and inflammatory mediators to the liver through the portal circulation, driving hepatic insulin resistance and steatosis. On this account visceral adipose is the source of the inflammatory burden, not its product.

This reframing costs the framework nothing and gains it a great deal. The two-signal model is unchanged. What changes is that visceral adipose becomes the compartment in which the two signals converge and from which their consequences propagate — which is a stronger and more specific claim than saying that removing the signals makes the compartment smaller.

7. A testable prediction: the reversibility asymmetry

The framework generates one prediction that is distinctive enough to be worth stating on its own, because few competing accounts make it and it can be checked in a modest study.

Antigen and antibody clear on entirely different timescales. Cell-surface glycans turn over continuously, so metabolically incorporated Neu5Gc should be substantially cleared within weeks of dietary elimination. Circulating anti-Neu5Gc IgG, by contrast, is maintained by plasma cells and decays over months to years.

Because pathology requires both the installed antigen and the circulating antibody, removing the antigen should produce clinical improvement well before antibody titres normalise. The framework therefore predicts a dissociation: symptomatic and biochemical improvement at four to eight weeks, with anti-Neu5Gc titres substantially unchanged at that point and falling only over subsequent months.

A framework in which improvement tracked titre decline would be a different framework, and would be falsified by this observation. A framework in which titres fell without clinical change would also be in trouble. The prediction is therefore genuinely discriminating, and it is cheap to test within an existing registry.

8. What this framework does not claim

The following objections are recorded because a reader will raise them. Three of the five are not answered, and are marked as such.

Objection	Status
Energy balance explains it	Not answered. Removing beef, bovine dairy and wheat alters energy density and diet quality substantially. The Armandi trial did not control energy intake, and its authors explicitly note that the observed BMI reduction may itself account for part of the metabolic improvement. Until an isocaloric design is run, energy balance remains the parsimonious explanation for any mass change.
IL-1β blockade does not shrink people	Conceded, and load-bearing. CANTOS suppressed IL-1β signalling for years and cut cardiovascular events without acting as a weight-loss agent. This is the single strongest reason the framework must not claim that inflammation drives fat mass. It is simultaneously the strongest support for the vascular claim, which requires no change in mass at all.
Coeliac patients gain weight on gluten-free diets	Partially answered. Weight gain after gluten withdrawal in coeliac disease follows restored mucosal absorption in a previously malabsorbing gut, and is frequently accompanied by a shift toward energy-dense processed substitutes. It does not bear on ATI signalling in an intact gut. The observation nonetheless demonstrates that removing wheat is not intrinsically slimming.
South Asian epidemiology	Not answered. Populations with low wheat and low bovine intake show high visceral adiposity at low BMI. Any account that makes ATIs or Neu5Gc necessary for visceral adiposity is falsified by this. The framework must therefore position both as modifiers of inflammatory tone, not as required causes of fat deposition.
The Cmah-null model is over-tuned	Acknowledged in the literature. Soulillou and colleagues argued in PNAS that the doubly deficient model amplifies the xenosialitis effect beyond what human physiology would support; the original authors accepted the general limits of extrapolation while defending the model's relevance. Tier B claims are stated at animal-model strength for exactly this reason.

9. Proposed studies

9.1 The isocaloric ATI trial (tests the mass claim)

The mass claim cannot be assessed by any design that permits energy intake to vary. The requirements are strict:

- Isocaloric arms with pair-fed equivalence, energy intake verified rather than self-reported.
- ATI-reduced versus ATI-containing diets matched for macronutrient composition and fibre.
- Visceral adipose tissue quantified by MRI at the L4–L5 level. Waist circumference and bioimpedance are inadequate and will not survive review.
- Twelve weeks minimum. Visceral adipose does not respond meaningfully over six.
- Hepatic fat fraction measured in parallel, on the expectation that it will move first and further.

A trial of this design that showed hepatic improvement without visceral fat loss would refute the mass claim while confirming the inflammatory-tone claim. Both outcomes are informative, which is the point.

9.2 The sialoglycan microarray (tests the epitope model)

The framework′s central technical proposition is that anti-Neu5Gc antibodies recognise extended glycan epitopes including the underlying chain, rather than the Neu5Gc residue in isolation. If correct, Neu5Gc content is the wrong measurement and epitope configuration is the right one. This predicts that individual antibody repertoires should map to individual dietary responses, including negative responses to species whose tissues contain abundant Neu5Gc.

A sialoglycan microarray run against individual sera would test this directly, and would do so before any dietary intervention. Establishing the antibody profile first, then predicting food responses from it, closes the placebo gap by design rather than by blinding — a prediction made in advance and recorded cannot be produced by expectation.

9.3 The titre dissociation study (tests the reversibility asymmetry)

Serial anti-Neu5Gc titres alongside symptom and inflammatory-marker tracking, at baseline, four weeks, eight weeks, six months and twelve months, in participants undertaking dietary elimination. The prediction stated in Section 7 is either observed or it is not.

10. Conclusion

The framework has a defensible claim about visceral adipose tissue and an undefended one, and the difference between them is the difference between a paper that survives review and one that does not.

Removing wheat ATIs and dietary Neu5Gc is proposed here as a means of lowering the inflammatory output of the mesenteric compartment, with consequences for hepatic fat, insulin sensitivity and endothelial health. The evidence for the ATI half of that proposition is established in humans. The evidence for the Neu5Gc half is a causal demonstration in a humanised animal model with an acknowledged critique attached. The convergence of the two in mesenteric adipose is a hypothesis, and the specific claim that either intervention reduces visceral fat mass is not supported by anything presently available.

The framework should be judged on the first claim. It is the stronger one, it is the one the evidence supports, and it does not require the second to be true.

References

1. Junker Y, Zeissig S, Kim SJ, et al. Wheat amylase trypsin inhibitors drive intestinal inflammation via activation of toll-like receptor 4. J Exp Med. 2012;209(13):2395–2408.

2. Ashfaq-Khan M, Aslam M, Qureshi MA, et al. Dietary wheat amylase trypsin inhibitors promote features of murine non-alcoholic fatty liver disease. Sci Rep. 2019;9:17463.

3. Armandi A, Bespaljko H, Mang A, Huber Y, Michel M, Labenz C, Galle PR, Neerukonda M, Bugianesi E, Schuppan D, Schattenberg JM. Short-term reduction of dietary gluten improves metabolic-dysfunction associated steatotic liver disease: a randomised, controlled proof-of-concept study. Aliment Pharmacol Ther. 2024;59(10):1212–1222.

4. Kawanishi K, Dhar C, Do R, Varki N, Gordts PLSM, Varki A. Human species-specific loss of CMP-N-acetylneuraminic acid hydroxylase enhances atherosclerosis via intrinsic and extrinsic mechanisms. Proc Natl Acad Sci USA. 2019;116(32):16036–16045.

5. Soulillou JP, et al. Can we extrapolate from a Cmah−/−Ldlr−/− mouse model a susceptibility for atherosclerosis in humans? Proc Natl Acad Sci USA. 2020;117(4):1845–1846.

6. Kawanishi K, Varki A, Gordts PLSM, et al. Reply to Soulillou et al.: difficulties in extrapolating from animal models exemplify unusual human atherosclerosis susceptibility and mechanisms via CMAH loss. Proc Natl Acad Sci USA. 2020;117(4):1847–1848.

7. Kawanishi K, Coker JK, Grunddal KV, Dhar C, Hsiao J, Zengler K, Varki N, Varki A, Gordts PLSM. Dietary Neu5Ac intervention protects against atherosclerosis associated with human-like Neu5Gc loss. Arterioscler Thromb Vasc Biol. 2021;41(11):2730–2739. doi:10.1161/ATVBAHA.120.315280

8. Padler-Karavani V, Yu H, Cao H, et al. Diversity in specificity, abundance and composition of anti-Neu5Gc antibodies in normal humans: potential implications for disease. Glycobiology. 2008;18(10):818–830.

9. Samraj AN, Pearce OMT, Läubli H, et al. A red meat-derived glycan promotes inflammation and cancer progression. Proc Natl Acad Sci USA. 2015;112(2):542–547.

10. Dhar C, Sasmal A, Varki A. From 'serum sickness' to 'xenosialitis': past, present and future significance of the non-human sialic acid Neu5Gc. Front Immunol. 2019;10:807.

11. Ridker PM, Everett BM, Thuren T, et al. Antiinflammatory therapy with canakinumab for atherosclerotic disease. N Engl J Med. 2017;377(12):1119–1131.

12. Guerrero-Flores GN, Pacheco FJ, Boskovic DS, Pacheco SOS, Zhang G, Fraser GE, Miles FL. Sialic acids Neu5Ac and KDN in adipose tissue samples from individuals following habitual vegetarian or non-vegetarian dietary patterns. Sci Rep. 2023;13. doi:10.1038/s41598-023-38102-z

This paper contains no clinical guidance and no dietary protocol. Nothing in it should be construed as medical advice.